

Exam 1 - CY1010 course (50 marks)

Date: 07.09.2023

Time: 2:15 pm – 3:45 pm

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1. What is the odorizer used to detect gas leakage in the CNG/LPG cylinders?

- a) H_2S b) Mercaptene ☒ c) Mercaptan d) Gasoline

Ans: ☒ C

2. The most potent greenhouse gas in terms of efficiency is

- a) N_2O b) Soot ☒ c) Perfluorocarbons d) CH_4

Ans: ☒ C

3. The source of second-generation biofuel is

- a) Sawdust b) Agriculture waste ☒ c) Lignocellulose ☒ d) All the above

Ans: ☒ C

4. Which statement is correct regarding the energy flow in the ecosystem?

- a) It follows the first law of thermodynamics, but not the second law
b) It doesn't follow the laws of thermodynamics
☒ c) It follows both the first and second law of thermodynamics
d) It follows the second law of thermodynamics, but not the first law

Ans: ☒ C

5. Which of the following is the most effective fuel for the nuclear fission process?

- ☒ a) ^{238}U b) ^{236}U ☒ c) ^{235}U d) 3H

Ans: ☒ C

6. Sunderlal Bahuguna is known for

- a) Ornithology b) Chipko movement c) Water conservation ☒ d) Green advocate

Ans: ☒ D

7. Which chemical production process consumes most of the CO_2 industrially?

- a) Salicylic acid ☒ b) Urea c) Polycarbonates d) Methanol

Ans: ☒ B

8. The examples of particulate matter causing global warming are

- a) Soot and CO_2 b) Sulfate and CH_4 ☒ c) Sulfate and N_2O d) Soot and sulfate

Ans: ☒ C

9. The molecule used for the functionalization of adsorbents for CO_2 capturing is

- a) Imine ☒ b) Amine c) Amide d) All the above

Ans: ☒ B

10. The blue hydrogen means

- a) It is produced from methane and carbon emissions are released into the atmosphere
b) It is produced from coal and carbon emissions are released into the atmosphere
☒ c) It is produced from methane and carbon emissions are captured/stored
d) It is produced from water and carbon emissions are captured/stored

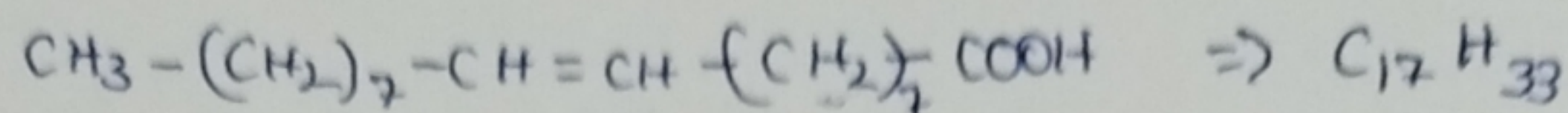
Ans: ☒ C

11. Which statement is correct?

- a) Methane will be released during rice cultivation under oxygen conditions
☒ b) Methane will be released during rice cultivation under oxygen-free conditions
c) Methane will not be released during rice cultivation
d) Both a and b

Ans: ☒ B

12. What is the most abundant element in the lithosphere?
 a) Al ☒ b) Si ☒ c) Ti ☐ d) Fe ☐
 Ans: ☒ b
13. The main components of LPG are
 a) Propene and butane ☒ b) Propane and butane ☒ c) Propane and butene ☐ d) Propene and butene ☐
 Ans: ☒ b
14. Biodiesel can be produced from
 a) Fats ☒ b) Oils ☒ c) Triglycerides ☐ d) All the above ☐
 Ans: ☒ b
15. How many neutrons are released during nuclear fission (U-235)?
 a) One ☒ b) Three ☒ c) Two ☐ d) Four ☐
 Ans: ☒ b
16. Rajender Singh is known for
 a) Ornithology ☒ b) Chipko movement ☒ c) Water conservation ☐ d) Green advocate ☐
 Ans: ☒ b
17. Which statement is correct?
 a) Both gasoline and diesel engines emit soot ☐
 b) Only gasoline engines emit soot ☐
 c) Only diesel engines emit soot ☒
 d) Both gasoline and diesel engines do not emit soot ☐
 Ans: ☒ c
18. The source of second-generation biofuel is
 a) Lignocellulose ☒ b) Lipids ☐ c) Fats ☐ d) All the above ☐
 Ans: ☒ a
19. Which fuel does not emit carbon emissions?
 a) CNG ☒ b) LPG ☐ c) Coal ☐ d) Hydrogen ☒
 Ans: ☒ d
20. Which reagent is used in the solvent-based direct air capture method of CO₂?
 a) CaO ☒ b) Amine ☐ c) KOH ☒ d) Toluene ☐
 Ans: ☒ c
21. Fuel cells
 a) They can use any kind of hydrogen to generate electricity and emit water ☒
 b) They can use green hydrogen only to generate electricity and emit water ☐
 c) They can use any kind of hydrogen to generate electricity, but both water and CO₂ emissions are possible ☐
 d) Both a and b ☐
 Ans: ☒ a
22. Which among the following compounds have the highest global warming potential?
 a) Nitrous oxide ☒ b) Sulfur hexafluoride ☒ c) Water vapour ☐ d) CFC ☐
 Ans: ☒ b
23. Which atmosphere is suitable for satellites?
 a) Thermosphere ☒ b) Mesosphere ☐ c) Exosphere ☐ d) All the above ☐
 Ans: ☒ a
24. Which among the following products can be produced via plastic waste recycling?
 a) Graphene ☒ b) Hydrocarbon fuels ☒ c) Monomers ☐ d) All the above ☐
 Ans: ☒ b
25. Which fuel emits higher carbon emissions?
 a) LPG ☒ b) CNG ☐ c) Petrol ☒ d) Biodiesel ☐
 Ans: ☒ c



26. The alkyl chain in triglycerides of olive oil is

- a) $n\text{-C}_{17}\text{H}_{35}$ ☒ b) $n\text{-C}_{17}\text{H}_{33}$ c) $n\text{-C}_{18}\text{H}_{37}$ d) $n\text{-C}_{19}\text{H}_{39}$

Ans:

27. Perfluorooctanesulfonic acid can protect fabrics from

- a) Water ☒ b) Oils c) a & b d) None

Ans: ~~B~~

28. Detergent does not contain

- a) Bleaches b) Builders c) Enzymes ☒ d) Polyoxides

Ans: ~~D~~

29. Identify the natural polymer from the following

- a) Vulcanized rubber ☒ b) Cellulose c) Glycerol d) Nylon-66

Ans: ~~B~~

30. From 1950 to 2015, how many million metric tons of polymers were discarded as a plastic waste globally

- a) 8300 ☒ b) 4900 c) 4600 d) 5600

Ans: ~~B~~

31. Globally, how much amount of plastic is being produced in one quarter of a year

- a) 10^9 kg ☒ b) 10^{11} kg c) 10^{12} kg d) 10^{10} kg

Ans: ~~B~~

32. In one of the States of India, pesticides listed in dirty dozen are being used for agricultural purposes. Identify the correct statement or statements from the following.

- a) These pesticides cannot cause harm to the entire World
☒ b) These pesticides can cause harm to the entire World
 c) These pesticides can cause harm only to that State
 d) These pesticides can cause harm only to India

Ans: ~~B~~

33. Which is a false statement about Polychlorobiphenyls?

- ☒ a) Volatilizes into the environment
 b) Soluble in fats
 c) Brings flexibility to the plastics as an additive
 d) Employed as a fluid in transformers

Ans: ~~A~~

34. The molar mass of a polymer is in the range of

- a) $10000 - 100000 \text{ g mol}^{-1}$ b) $10 - 1000 \text{ Da}$ ☒ c) $100 - 1000 \text{ Da}$ d) $10000 - 100000 \text{ Da}$

Ans: ~~C~~

35. Which of the following polymer products are quickly discarded as a waste?

- ☒ a) Packaging material
 b) Building and construction material
 c) Textiles
 d) Transportation material

Ans: ~~A~~

36. Which of the following organic chemical does not contain carbon-chlorine bonds in the molecular structure?

- a) Aldrin b) Dieldrin c) DDT ☒ d) Latex

Ans: ~~D~~

37. To be highly persistent in the environment, half-life of a chemical in the sediment must be.

- ☒ a) > 6 months b) ≥ 4 months c) ≥ 2 months d) ≥ 1 month

Ans: ~~A~~

38. Identify a natural pesticide from the following.

- ☒ a) Garlic oil b) Endosulfan c) Lindane d) Toxaphene

Ans: ~~A~~

39. What is the best solution to plastic pollution?

a) Recycle

☒ b) Reduce

c) Reuse

d) Incineration

Ans: B

40. How much percentage of plastic waste remain in the environment at global level as per the past decade statistics?

a) 18%

b) 68%

c) 48%

☒ d) 58%

Ans: D

41. What is the function of electrostatic precipitators in modern incinerators?

a) Separates metals

☒ b) Regulates particulate matter

c) Supplies energy to the steam boiler

d) Stores electricity that is produced

Ans: B

42. Cleansing action of soaps and synthetic surfactants is due to

a) Hydrophilicity

b) Lipophilicity

☒ c) Amphiphilicity

d) Hydrophobicity

Ans: C

43. What is the cause for the algal blooms formation in the lakes

a) Sulfates

b) Carboxylates

☒ c) Phosphates

d) Sulfonates

Ans: C

44. Single-use plastics manufactured in the year 2023 would degrade naturally by

a) 2033

b) 2123

☒ c) 3023

d) 2073

Ans: C

45. Higher the Octanol/water partition coefficient (K_{ow}) of a chemical

☒ a) Lesser the tendency to dissolve in water

b) Lesser the tendency to dissolve in the fat

c) Lesser the tendency to bioaccumulate

d) Lesser the harmful impact

Ans: A

46. Failure of the potato crop in Ireland in 1845 is due to

☒ a) Fleas

b) Rodents

c) Fungus

d) Bacterium Yersinia pestis

Ans: A

47. Nylon-66 is made by

a) Addition polymerization

b) Crosslinking polymerization

☒ c) Condensation polymerization

d) Co-polymerization

Ans: C

48. Ca^{+2} and Mg^{+2} ions are known to precipitate out upon interaction with

a) Long chain carboxylic acids

☒ b) Long chain carboxylates

c) Long chain sulfonates

d) Long chain sulfonic acids

Ans: B

49. Minamata disease was caused by

a) Mercury sulfate

b) Methyl mercury

☒ c) Polychlorinated biphenyls

d) Dioxins

Ans: C

50. PBT stands for

a) Personal Best Time

b) Performance Based Training

c) Profit Before Tax

☒ d) Persistent, Bioaccumulative and Toxic

Ans: D